# Reliable and Trustworthy Artificial Intelligence

## Assignment #3

Sang-Ki Ko

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**Instructions**

- **Release date:** Wednesday, April 29, 2026
- **Due date:** Wednesday, May 13, 2026, by 11:59 PM
- **Submission:** Upload the project to GitHub and submit the link to Uclass. Late submissions will not be accepted, either through the system or email.

> **Important:**
>
> - Reproducibility is crucial for grading. Your GitHub project should include a `requirements.txt` (or a Dockerfile) that describes all external modules and environment setup.
> - Please include comments in your code to help me understand it.
> - Include a `test.py` file that demonstrates running Marabou on your chosen model. The specific format is not required, but it should demonstrate your code's validity.

## 1 Overview

In this assignment, you will familiarize yourself with **Marabou**, a Satisfiability Modulo Theories (SMT) based tool for neural network verification. Neural network verification aims to formally prove properties about a network's behavior — for example, that small perturbations to an input cannot change the predicted class. You will learn how to set up Marabou, explore its built-in resources, and run it on an external model and dataset of your choosing.

> **Reference:** Marabou was introduced in the paper "The Marabou Framework for Verification and Analysis of Deep Neural Networks" (Katz et al., CAV 2019). Reading the paper's introduction will help you understand the tool's approach to verification.

# 2 Problem 1: Explore the Marabou Resources Directory

## Background

Marabou provides a collection of sample models and datasets in its repository. Understanding what is available will help you choose an appropriate external model for the next part.

## What to do

1. Visit the Marabou GitHub repository: `https://github.com/NeuralNetworkVerification/Marabou`
2. Navigate to the resources directory:
   `https://github.com/NeuralNetworkVerification/Marabou/tree/master/resources`
3. Review the files available in this directory and document:
   - What types of models are provided (architecture, format)?
   - What datasets or input specifications are included?
   - What example verification queries are available?

> **Tip:** Pay attention to the supported model formats (e.g., `.nnet`, `.onnx`, `.pb`). This will be important when selecting your own model in Problem 2.

# 3 Problem 2: Run Marabou with an External Model and Dataset

## Background

To demonstrate your understanding of Marabou, you will apply it to a model that is *not* already included in the resources directory. This requires understanding Marabou's input format, formulating a verification query, and interpreting the results.

## What to do

Choose an external model and dataset, set up Marabou, and run a verification query.

### Step-by-step guide

1. **Install Marabou.**
   - Clone the repository and follow the build instructions in the README.
   - Verify the installation by running one of the provided examples.
   - Document any issues encountered during installation and how you resolved them.
2. **Select an external model and dataset.**
   - Choose a model that is *not* already in the Marabou resources directory.
   - The model should be small enough for Marabou to handle — very large models (e.g., full-size ResNets) will likely time out. A simple fully connected network or small CNN is recommended.
   - Ensure the model is in a format supported by Marabou (e.g., `.onnx`, `.nnet`, or `.pb`).
3. **Formulate a verification query.**
   - Define input constraints (e.g., an $\ell_\infty$-ball of radius $\varepsilon$ around a specific input).
   - Define output constraints (e.g., the network should always predict the same class within the perturbation region).
   - Example: "For input image $x$ classified as digit 3, verify that all inputs within $\|x' - x\|_\infty \leq 0.01$ are also classified as digit 3."
4. **Run Marabou and interpret results.**
   - Run the verification query using Marabou's Python API or command-line interface.
   - Record whether the property was verified (`UNSAT`) or a counterexample was found (`SAT`).
   - If a counterexample is found, visualize or describe the adversarial input.
   - Report the verification time and any resource usage observations.

> **Tip:** Start with a very small $\varepsilon$ value and a simple model. If verification takes too long, try reducing the model size or the perturbation radius. Marabou's Python interface (`maraboupy`) is often the easiest way to get started — see the documentation and Jupyter notebook examples in the repository.

# 4 Deliverables & Grading

## 4.1 Exploration report (20%)

A summary of the Marabou resources directory: what models, datasets, and example queries are available.

### 4.2 Implementation & results (50%)

1. Your code for loading the external model, formulating the verification query, and running Marabou.
2. A clear record of:
   - Which model and dataset you chose, and why.
   - Any modifications or preprocessing steps required for compatibility.
   - The verification result (`SAT/UNSAT`), runtime, and any counterexamples found.

### 4.3 Report (30%)

Submit a short PDF report (**1–2 pages**) that includes:

1. A description of your model, dataset, and verification query.
2. The results and their interpretation.
3. A brief discussion of Marabou's strengths and limitations based on your experience (e.g., scalability, ease of use, supported model formats).

**On the use of AI tools**

You may use AI assistants (e.g., ChatGPT, GitHub Copilot) as a *learning aid*, but you must demonstrate genuine understanding of your work. To ensure this:

- Your GitHub repository must show a **meaningful commit history** that reflects incremental development — not a single bulk commit. We will inspect the git log.
- The report must include your **own interpretation and reasoning**, not generic descriptions. Discuss specific behaviors you observed in your experiments, including any unexpected results.

## 5 Submission Guidelines

**Submission Checklist**

Upload the following to GitHub and submit the repository link to Uclass:

1. `requirements.txt` (or `Dockerfile`) — environment setup and dependencies.
2. `test.py` — runs Marabou on your model and produces verification results.
3. `report.pdf` — your analysis report (1–2 pages).
4. `README.md` — brief instructions on how to install Marabou and run your code.